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How well do global ocean approaches constrain local fCO_2?

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Abstract

The ocean absorbs 29% of humanity's annual anthropogenic carbon dioxide (CO_2) emissions, and the future of climate change is strongly dependent on how this sink evolves. Marine Carbon Dioxide Removal (mCDR) approaches to enhance this sink are actively being developed. In the interest of understanding how well state-of-the-art global products and models can help to distinguish mCDR signals from the background ocean carbon sink, we use sparse independent data to quantify their local-scale ($1° \times 1°$, monthly) errors in CO_2 fugacity (fCO_2), the primary regulator of air-sea CO_2 fluxes. Our 2000–2023 analysis uses 5 large regions ('superbiomes') to aggregate data across high-latitude, subtropical and equatorial zones, and we compare to 10 products and 10 models. The observed long-term trend of surface ocean CO_2 due to rising atmospheric CO_2 is consistently estimated. At the same time, we find substantial fCO_2 biases for the ensemble means: ± 5 μatm for the products, and from $+1$ to $+15$ μatm for the models. Across the superbiomes, ensemble-mean unbiased root-mean-square errors (uRMSEs) are 19-37 μatm for the products and 25–56 μatm for models. Individual products and models have larger errors. Seasonality is well-correlated to the independent data for the products and for the models in the subtropics, but the largest component of uRMSE also lies at this timescale. Deseasonalized variability has smaller uRMSEs and low correlations to independent data. In summary, local-scale errors in global observation-based products and models for the background ocean carbon sink are more than 1 order of magnitude larger than the local to regional-scale signals likely for open-ocean mCDR ($\sim$1 μatm). Reducing local-scale uncertainty in the ocean carbon sink will be critical if these products and models will be part of mCDR monitoring, reporting and verification systems. Improving the representation of seasonality should be a key target for improving both observation-based products and models.

1. Introduction

Given the critical importance of the ocean's uptake of anthropogenic carbon to the future of the global carbon budget and climate change (Canadell *et al* 2021, Gruber *et al* 2023, McKinley *et al* 2023, McKinley 2026) it is important to understand the accuracy of current estimates of the sink. This sink occurs due to rising atmospheric CO_2 and has taken up 29% of annual anthropogenic CO_2 emissions in the last decade (Friedlingstein *et al* 2026a).

If marine Carbon Dioxide Removal (mCDR) approaches will be implemented, the background sink must be quantified as a counterfactual against which additional uptake occuring as a result of the mCDR technology ('additionality') can be evaluated. In this paper and for consistency with the significant literature on this topic, we will refer to the ocean's uptake of anthropogenic carbon that occurs without direct intervention as the *ocean carbon sink*. Engineered approaches will be explicitly identified as mCDR (NASEM 2022).

When globally-integrated and annually-averaged, the Global Carbon Budget 2025 (Friedlingstein *et al* 2026a) estimates that the ocean carbon sink was 3.2 ± 0.4 PgC yr^{-1} for 2014-2024. This uncertainty has a magnitude of 13% (1σ) of the mean flux for the global integral. Various analyses have shown that

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Table 1. Observation-based Products (top) and Global Ocean Biogeochemical Models (bottom) in this study.

Product	Reference
CMEMS-LSCE-FFNNv2	Chau *et al* (2022)
CSIR-ML6	Gregor *et al* (2019)
Jena-MLS	Rödenbeck *et al* (2022)
JMA-MLR	Iida *et al* (2021)
LDEO-HPD	Gloege *et al* (2022), Bennington *et al* (2022b)
NIES-ML3	Zeng *et al* 2022
OceanSODA-ETHZv2	Gregor *et al* (2024)
UExP-FNN-U	Watson *et al* (2020), Ford *et al* (2024)
VLIZ-SOMFFN	Landschützer *et al* (2016, 2020)
LDEO-Residual	Bennington *et al* (2022b)
ACCESS CSIRO	Law *et al* (2017)
CESM-ETHZ	Doney *et al* (2009)
CNRM	Berthet *et al* (2019), Séférian *et al* (2019)
FESOM-2.1-REcoM3	Gurses *et al* 2023
MOM6-COBALT (Princeton)	Liao *et al* (2020)
MPIOM-HAMOCC6	Lacroix *et al* (2020)
MRI-ESM2-3	Tsujino *et al* (2024), Sakamoto *et al* (2023)
NEMO4.2-PISCES (IPSL)	Aumont *et al* (2015)
NEMO-PlankTOM12	Wright *et al* (2021)
NorESM1-OCv1.2	Schwinger *et al* (2016)

when global flux estimates are disaggregated into large regions spanning one or several ocean gyres, differences between the estimates are much larger than 13% for both the long-term mean and seasonality (McKinley *et al* 2016, Fay and McKinley 2021, DeVries *et al* 2023, Rodgers *et al* 2023, Terhaar *et al* 2024). Not yet reported in the literature are uncertainties at the smallest scales ($1° \times 1°$, monthly) of the Global Ocean Biogeochemical Models (GOBMs, or 'models') and observation-based products (or 'products') used in the Global Carbon Budget. This paper addresses this gap.

Currently, commercial, non-profit and academic research seeks to develop mCDR technologies and the observational and modeling approaches that may be used to quantify the real-world impacts of mCDR deployments (NASEM 2022). What will be the baseline estimate of the ocean carbon sink against which perturbations due to mCDR interventions will be measured? Given that the globally-integrated ocean carbon sink is six orders of magnitude larger than all currently-verifiable novel CDR, and that the uncertainty of this sink is also five orders of magnitude larger (Powis *et al* 2023, Friedlingstein *et al* 2026a), robust estimates of the ocean carbon sink are essential if the very small perturbations to the air-sea fluxes expected from mCDR are to be detected (Wang *et al* 2022, Zhou *et al* 2024). This need applies both for the project-scale where a counterfactual for an ocean without mCDR intervention will be needed, as well as for the global scale where net impacts on the atmospheric CO_2 concentration will need to be assessed (Lamarque *et al* 2026).

For large oceanic regions, differences across observation-based product and GOBM ensembles of the GCB are large (McKinley *et al* 2016, Fay and McKinley 2021, Terhaar *et al* 2024). Friedlingstein *et al* (2026a) compare observation-based products to independent data from GLODAP (Lauvset *et al* 2016, 2021, Olsen *et al* 2016) and SOCAT Flag E (quality of ± 10 μatm) (Bakker *et al* 2016). For the globe and for North, Tropical and South regions separated at 30°N and 30°S, they find biases are generally within 10 μatm and root mean square errors (RMSEs) of 20–50 μatm. The analysis presented here expands these comparisons to include additional independent data, adds GOBMs for equivalent comparisons, and evaluates the dominant timescales of errors for both products and GOBMs. This work improves understanding of the underlying sources of error in ocean carbon sink estimates from the Global Carbon Budget. It is also a first step in understanding the extent to which these global products could be used to quantify the ocean carbon sink that would have occurred in a given location without implementation of mCDR; and thus to be used as the basis for calculations of mCDR additionality.

2. Methods

2.1. Observation-based products and global ocean biogeochemical models

Here, we evaluate the observation-based products and Global Ocean Biogeochemical Models (GOBMs) (table 1) used in GCB2024 (Friedlingstein *et al* 2025) for 2000–2023. The Global Carbon Budget (GCB) protocol allows only one submission from each research group. Our Lamont-Doherty Earth Observatory (LDEO) Ocean Carbon research group

maintains two products, LDEO-Hybrid Data Physics (LDEO-HPD) (Gloege *et al* 2022, Bennington *et al* 2022b) and LDEO-Residual (Bennington *et al* 2022a, Heimdal *et al* 2026), with LDEO-HPD being used in the GCB. Here we include both the LDEO-HPD and LDEO-Residual products. Observation-based products are all trained with SOCAT (Bakker *et al* 2016), and thus these data will be excluded from all comparisons to the observation-based products. Per the GCB protocol, observation-based products and GOBMs report fCO_2.

2.2. Independent data

We utilize the merged Global Ocean Data Analysis Project (GLODAP) v2.2023 dataset (Lauvset *et al* 2016, 2021, Olsen *et al* 2016) (downloaded on July 19, 2024). GLODAP contains quality-controlled profiles of various carbonate system parameters with depth, including DIC, Total Alkalinity (TA), as well as temperature and salinity and a variety of other ocean parameters. While fCO_2 is available in the v2.2023 version, it is not quality controlled and there is limited data available for that variable. Therefore, we calculate pCO_2 from surface ($\leqslant$10 m) bottle observations using the CBSYST python package (cbsyst-0.4.10). Inputs to the pCO_2 calculation are GLODAP values for pH, DIC, temperature, salinity, pressure, phosphate, and silicate and a constant average boron concentration of 415.7 μmol kg^{-1}. The resulting pCO_2 are converted to fCO_2 according to (Dickson *et al* 2007). Uncertainty is estimated at $\pm$12 μatm (Friedlingstein *et al* 2026a).

The LDEO Global Ocean Surface Water Partial Pressure of CO_2 Database, developed by Taro Takahashi, compiles surface water pCO_2 measurements to understand the ocean's role in the global carbon cycle (Takahashi *et al* 2019). The 2019 release includes about 14.2 million measurements from 1957 to 2019. The dataset includes only those observations measured using equilibrator-CO_2 analyzer systems, and have been quality-controlled based upon the stability of the system performance, the reliability of calibrations for CO_2 analysis and the internal consistency of data. We convert the reported pCO_2 to fCO_2 (Dickson *et al* 2007). Uncertainty on the LDEO dataset is estimated at $\pm$2.5 μatm (Fay *et al* 2024).

In the main text, the SOCAT fCO_2 database (Bakker *et al* 2016) is used only to remove any points that are also found in the LDEO database. This ensures that no data used to train the observation-based products is included in the independent dataset. In the Supplementary Material, the GOBMs are compared to an alternative merged dataset that includes SOCAT.

2.2.1. Combination of datasets

The independent surface fCO_2 dataset to which we compare the products and GOBMs is a combined set from GLODAP and the LDEO database, after SOCAT points are removed. This dataset is at 1° $\times$ 1° spatial and monthly temporal resolution for 2000-2023. We report the number of points for each of the 5 super-biomes (Fay and McKinley 2014, 2021) that are the focus for our analysis (table 2). Due to the extreme sparsity of the independent data, far-northern and southern ice biomes are not considered in this analysis. For the purpose of discussing the datasets in this section, 'global' is the union of the 14 non-ice open-ocean biomes, which leaves out coastal zones and marginal seas (Fay and McKinley 2014). The timeframe is 2000-2023.

To construct the final dataset, we applied the following data screening and merging steps. First, we remove any outlier points defined as those with fCO_2 higher than 650 μatm or lower than 100 μatm. This removes 0.7% from GLODAP, 0.02% from SOCAT, 0.02% from LDEO. Second, all points that are found in both the LDEO and SOCAT database are identified. These points are removed. This makes the 'LDEOnoSOCAT' dataset (N = 11 573, global). Third, to combine with surface ocean GLODAP data (N = 1043, global), we identify any 1° $\times$ 1° monthly points in both GLODAP and 'LDEOnoSOCAT' (N = 19) and these points are averaged. The remaining 1° $\times$ 1° monthly points in the final dataset are taken from either GLODAP or 'LDEOnoSOCAT'. This creates the 'GLODAP+LDEOnoSOCAT' surface ocean dataset that is our primary basis for comparison (N = 12 590, global). The final point counts per super-biome are provided in table 2.

Measurement uncertainty for the combined GLODAP+LDEOnoSOCAT dataset is estimated at 4.2 μatm using the square root of the weighted sum of squares, with the weighting being the percentage of data from GLODAP (8%) and LDEOnoSOCAT (92%). At the 19 monthly crossover points, the difference between LDEOnoSOCAT and GLODAP points is 1.9 μatm, within measurement error, and the uRMSE is 9.9 μatm.

In the Supplementary Material, we compare the GOBMs to an alternative dataset where SOCAT points are not removed. This alternative merged dataset is 23 times larger than GLODAP+LDEOnoSOCAT, but cannot be used for our primary comparisons because the observation-based products are trained with the SOCAT data.

2.3. Comparing to independent data

The sparse surface ocean independent data in GLODAP+LDEOnoSOCAT are used at 1° $\times$ 1° spatial and monthly temporal resolution. These points are extracted individually from each observation-based product and GOBM. We apply the removal of outliers procedure as applied to the independent data (section 2.2.1) in which we remove any outlier points from the products and GOBMs with fCO_2 lower than 100 μatm or higher than 650 μatm.

Table 2. Superbiomes and the number of independent datapoints in GLODAP+LDEOnoSOCAT for 2000-2023. Superbiomes are visualized in the inset maps in figures 4 and 5. The original biomes of (Fay and McKinley 2014) that are merged to create each superbiome are also indicated.

Superbiome	Biomes	Datapoints
North Seasonally Stratified (N.SS)	2,3,9,10	3892
North Subtropical Permanently Stratified (N.STPS)	4,11	3557
Equatorial (Eq)	5,6,12	1374
South Subtropical Permanently Stratified (S.STPS)	7,13,14	1723
Southern Ocean (S.Ocean)	15,16	1242

Both independent data and the points extracted from the products or GOBMs are aggregated over 'superbiomes' (table 2) using an area-weighted average. This leads to a timeseries ('All') for each superbiome for the independent data, for each product and for each GOBM. For a consistent approach to creation of the seasonal cycle, the trend of atmospheric xCO_2 from NOAA ESRL global marine boundary layer product is used to detrend each timeseries (Lan *et al* 2024) before averaging all data in each month ('Seasonal'). Separately, a linear trend is calculated from each 'All' timeseries, and this is removed along with the repeating seasonality to identify the detrended non-seasonal variability ('Variability').

2.4. Statistical measures of skill

Bias is calculated as the mean of the Prediction minus the mean of the Observation (bias $= \overline{P} - \overline{O}$). The sign of the bias indicates whether the Prediction (product or GOBM) overestimates (positive) or underestimates (negative) fCO_2.

Unbiased root mean squared error (uRMSE) measures the magnitude of the predicted error after the bias is removed. uRMSE penalizes larger errors and outliers. It is calculated as the square root of the mean of the squared unbiased error ($uRMSE = \frac{1}{N}\sqrt{\overline{[(P - bias) - O]^2}}$, where N is the number of data points being compared).

The Pearson correlation coefficient (r) measures how much the product or GOBM varies consistently with the independent data. Values near +1 indicate high tendency to vary together. It is calculated as the covariance between the predictions and the observations, divided by the product of their individual standard deviations.

All statistical analyses are done for each individual member of the 10 member ensembles for each products and GOBMs (tables 1 and 2). The presentation focuses on the ensemble mean and spread (1σ) of each metric for the suite of products or GOBMs in each superbiome.

3. Results

The large spread in surface fCO_2 estimates across observation-based products and GOBMs for a single month is demonstrated by the 1σ spread of the 10 ensemble members for January and July of 2015 (figure 1). For both months, the observation-based products are more consistent in their estimates across most of the ocean, with 1σ values as low as a few μatm in the central subtropical gyres of the Northern Hemisphere and <10 μatm in the central subtropical gyres of the Southern Hemisphere. In the far eastern equatorial Pacific, in the Arctic, and along the coasts, the 1σ spread exceeds 20 μatm in the products. The winter hemisphere tends to have a reduced spread compared to the summer hemisphere. The spread across the GOBMs is greater everywhere, exceeding 10 μatm in the central subtropical gyres of both hemispheres and exceeding 25 μatm over large swaths of the ocean, particularly in high latitude summer. The Arctic and coastal regions of GOBMs generally have the largest 1σ spread, exceeding 25 μatm. In summary, we find a lack of agreement among ensemble members at monthly temporal and $1° \times 1°$ spatial resolution for both observation-based products and GOBMs, with the degree of disagreement across the ensemble being greater for GOBMs.

Our goal here is to assess the skill of the observation–based products and GOBMs using co–located independent data. In figure 2, the sparsity of these data is illustrated with the 2000–2023 mean of one product differenced from the GLODAP+LDEOnoSOCAT independent data (figure 2(a)) and of one GOBM differenced from the same (figure 2(b)). Independent data are more numerous in the Northern Hemisphere and tropical Pacific. The Indian Ocean and the Southern Hemisphere are particularly sparse.

Our treatment of the data, described in section 2.3, is illustrated using the Northern Seasonally Stratified (N.SS) superbiome in figure 3. The individual observation-based products and GOBMs are sampled for the same month and $1° \times 1°$ locations where independent data exist (figure 2). For each superbiome, these data are collapsed to a single timeseries (figures 3(a) and (d)). A linear trend is calculated from the full timeseries (table 3). A seasonal climatology is created from the superbiome timeseries (figures 3(b) and (e)). The detrended non-seasonal variability (figures 3(c)

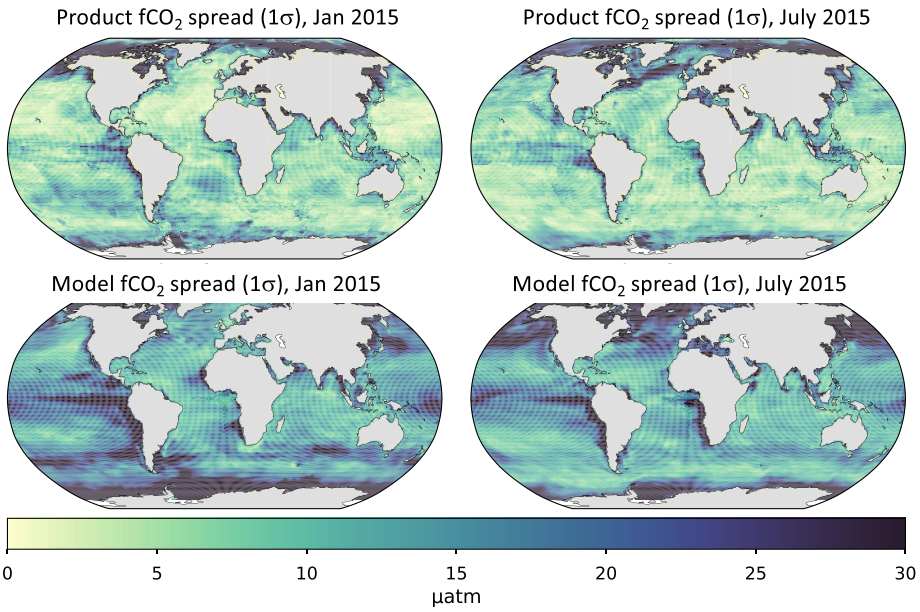


Figure 1. Spread of products and models in January and July 2015.

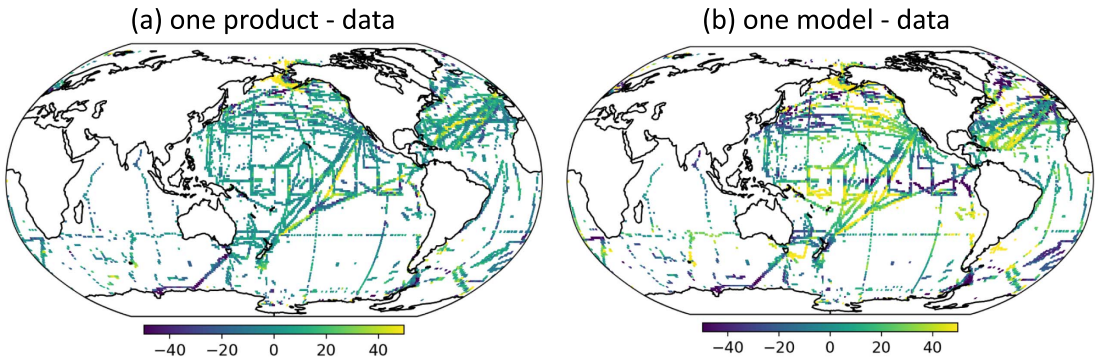


Figure 2. Example surface fCO_2 difference (in μatm) of (a) one product (SOM-FFN) and (b) one model (FESOM-RECOM) from the GLODAP+LDEOnoSOCAT independent data, mean for 2000-2023. Positive (negative) values indicate high (low) bias relative to observations.

and (f)) is the remainder after the linear trend and annually-repeating climatology are removed.

In the N.SS, independent data exist for most months prior to 2015 and then become more sparse (figures 3(a) and (d)). Visual comparison of the seasonality (figures 3(b) and (e)) indicates that products capture the independent observations with far greater fidelity than the GOBMs. It is not possible to visually assess skill with respect to the variability from figures 3(c) and (f).

Trends for the independent data and for the products and the GOBMs are nearly identical, ranging from 2.0-2.2 μatm/yr across the superbiomes (table 3). The strongly forced response of ocean fCO_2 to rapidly rising atmospheric CO_2 concentrations (McKinley *et al* 2020, DeVries *et al* 2023) is a consistent signal across the independent datasets, observation-based products, and GOBMs for all superbiomes.

For quantitative comparisons of skill against GLODAP+LDEOnoSOCAT, we turn to figures 4 and 5. Greater skill is indicated by lower bias, lower RMSE and a higher correlation coefficient (r). For observation-based products, the ensemble mean bias ranges from -3 to $+6$ μatm, while for the GOBMs, ensemble mean bias ranges from $+1$ to $+13$ μatm (figure 4). The full spread of GOBM bias is large compared to the products. This is a main difference between the products and the GOBMs. Bias in individual products ranges from value of -4 to $+9$ μatm, while GOBM bias spans a much larger range, from -18 to $+32$ μatm. In contrast to the GOBMs, the products have bias tightly clustered within a few μatm of the ensemble mean.

The highest uRMSE occurs in the N.SS superbiome (figure 5(a)). For the full timeseries ('All'), the uRMSE for products ranges from 30–45 μatm, with an ensemble mean of 37 μatm. For the GOBMs,

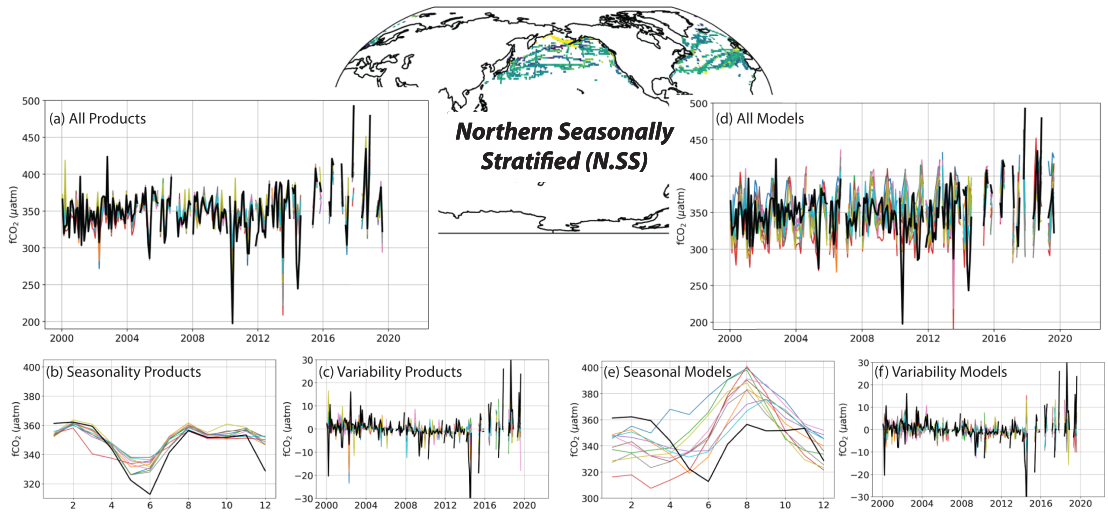


Figure 3. Visualization of the analysis comparing independent data to products and models sampled at the same points for the Northern Seasonally Stratified (N.SS) superbiome (map at top). (a) All products (colors) and data (black), (b) seasonality comparison for products, (c) detrended variability comparison for products; (d), (e), and (f) are the same, respectively, for the models. See text for further description. See figure S4 for color legend for the products and models.

Table 3. fCO_2 trends for 2000–2023 (μatm/yr) for each superbiome region and for the independent dataset GLODAP+LDEOnoSOCAT. For the observation-based products and the GOBMs, the ensemble mean $\pm$ the ensemble standard deviation are reported.

Superbiome	GLODAP+LDEOnoSOCAT	Products	GOBMs
N.SS	2.2	2.1 ± 0.03	2.1 ± 0.03
N.STPS	2.0	2.1 ± 0.01	2.1 ± 0.01
Eq	2.0	2.0 ± 0.04	2.0 ± 0.05
S.STPS	2.1	2.1 ± 0.01	2.1 ± 0.02
S.Ocean	2.0	2.0 ± 0.05	2.1 ± 0.03

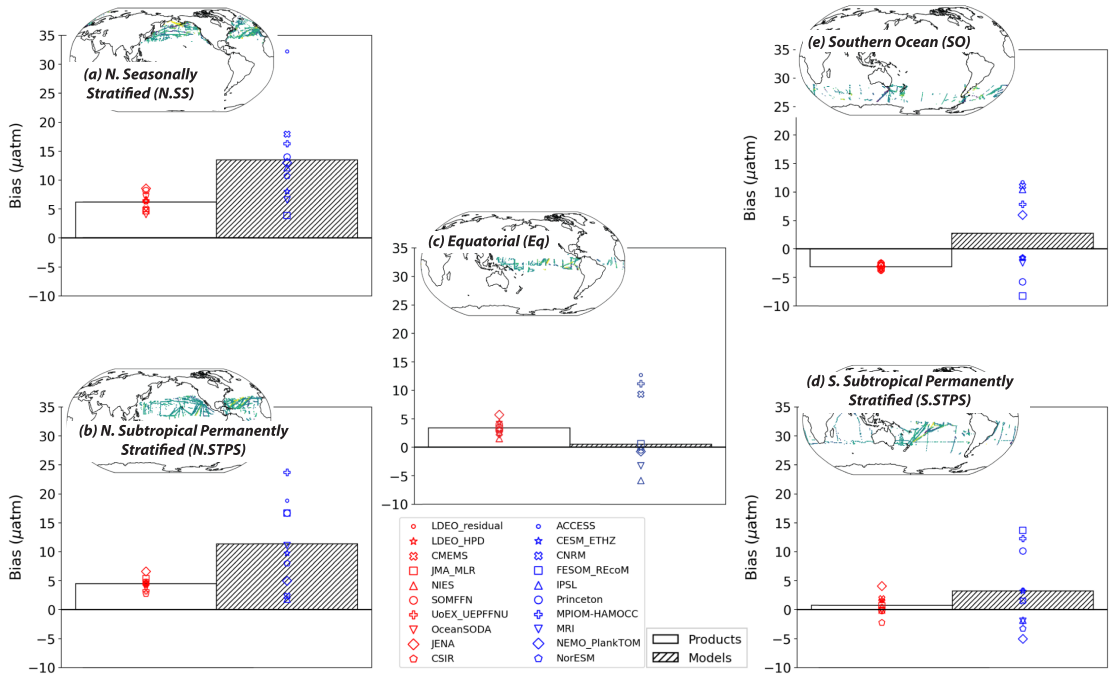


Figure 4. Bias against GLODAP+LDEOnoSOCAT independent data. Products are the open bars with red symbols, models are the hatched bars with blue symbols. NorESM has an extreme negative bias (-17.7 μatm) in the Equatorial biome that is not shown to improve the figure's overall appearance.

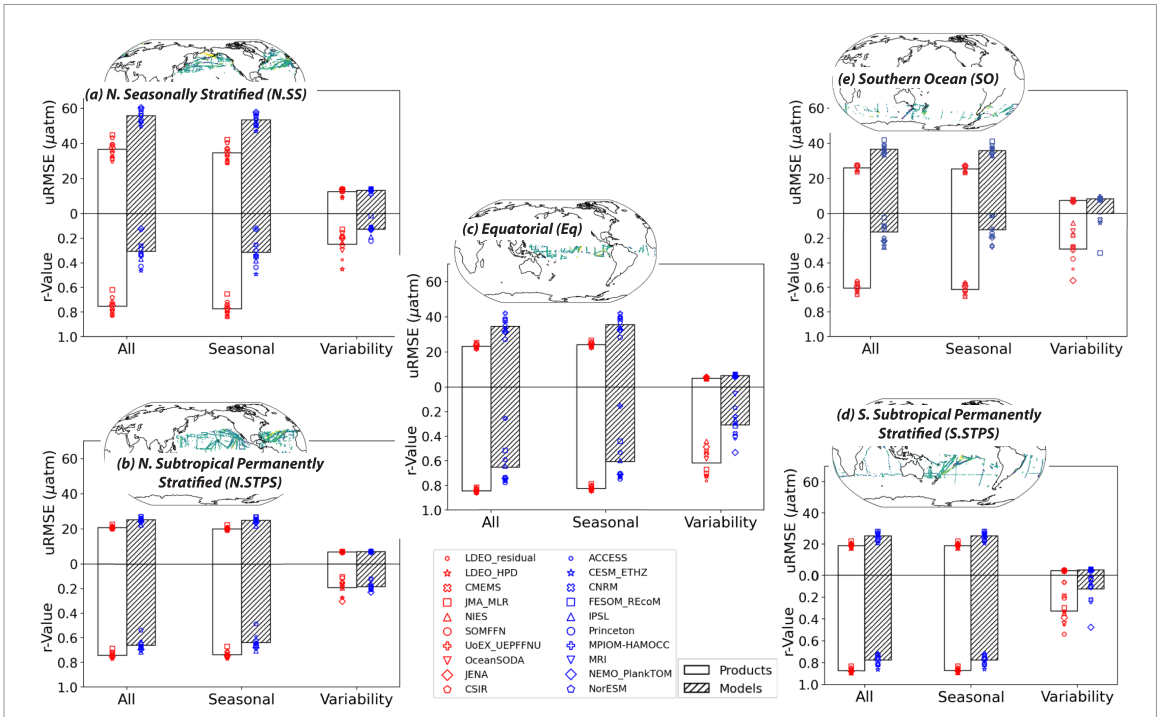


Figure 5. Unbiased Root-Mean-Square Error (uRMSE) and Correlation (r) against GLODAP+LDEOnoSOCAT independent data. In each subpanel, the first set of bars is for the full dataset (All), the second for the seasonality, and the third for the remaining non-seasonal variability (section 2.3 and figure 3). The top half of each panel shows uRMSE and the bottom shows correlation. Products are the open bars with red symbols, models are the hatched bars with blue symbols. If the ensemble-mean correlation is negative, no bar is shown.

the uRMSE range is 50–60 μatm with an ensemble mean of 57 μatm for GOBMs). The smallest uRMSEs are found in the two subtropical superbiomes (figures 5(b) and (d)). For each ensemble member, the uRMSE for the products is always closer to the ensemble mean than for the GOBMs. Ensemble-mean correlations for the products are also always higher than for the GOBMs. Notably, in the high-latitude superbiomes (N.SS and SO), the ensemble-mean correlation for the products exceeds 0.6, while the ensemble-mean correlation for the GOBMs is always 0.3 or less.

In the subtropics, ensemble-mean correlations for the full timeseries are reasonably high for both GOBMs and products: >0.75 for the Southern Subtropical Permanently Stratified (S.STPS, figure 5(d), and >0.65 in the Northern Subtropical Permanently Stratified (N.STPS, figure 5(b). In these superbiomes, the ensemble-mean uRMSE is ∼20 μatm for the products and ∼25 μatm for the GOBMs.

In the equatorial superbiome, when considering the full timeseries, the products all perform better than even the best performing GOBMs in terms of both uRMSE and correlation (figure 5(c)). The products have a mean uRMSE of 23 μatm and mean correlation of 0.85, while the GOBMs have a mean uRMSE of 35 μatm and mean correlation of 0.65.

In all superbiomes, most of the uRMSE and most of the correlation comes from the biome-aggregated seasonal cycle. This is shown with the central pair of bars ('Seasonal') in each sub-panel of figure 5 being only modestly smaller than the left pair ('All'). This finding is consistent with seasonality being the main signal of variation in surface ocean fCO_2 (Takahashi *et al* 1993), a signal captured by both the products and the GOBMs (figure 3), though with substantial residual errors against these independent data (figure 5).

The detrended non-seasonal variability has smaller mean uRMSEs (<13 μatm) and low ensemble-mean correlations <0.3 in all superbiomes, except the equatorial (figure 5). Ensemble-mean correlations for the products are higher than for the GOBMs, with the exception of N.STPS, where they are indistinguishable. In N.STPS and N.SS, the spread of the individual product correlations is larger than the spread of the GOBMs, while in equation, the opposite is true. In S.STPS and SO, the spread across products and across GOBMs are comparable.

These comparisons use GLODAP+LDEOnoSOCAT as independent data, allowing the relative skill of the products and GOBMs to be directly compared. However, the SOCAT database offers a much larger dataset from which the GOBMs are independent. In the Supplementary Material, comparisons for the GOBMs are repeated with the inclusion of the SOCAT data (Bakker *et al* 2016). Qualitatively, the comparisons are similar, but there are quantitative differences. Against GLODAP+LDEO+SOCAT, the GOBMs have similar biases and slightly improved performance for the full timeseries and seasonality in terms of uRMSE and r. For the variability, there are modestly higher uRMSEs and higher correlations

against GLODAP+LDEO+SOCAT. Please see the Supplementary Material for additional information.

4. Discussion and conclusion

These comparisons to independent fCO_2 data demonstrate that the GOBM ensemble is less skillful than the product ensemble in all superbiomes. For both products and GOBMs, biases are often larger than the estimated uncertainty of the independent dataset ($\pm$4.2 μatm), and uRMSEs are always larger for the full timeseries ('All') and the dominant timescale ('Seasonal'). Another important finding is that individual products tend to group together around the product ensemble mean much more tightly than the individual GOBMs around their ensemble mean. In other words, the products are more consistent in their skill than the GOBMs.

In all 5 superbiomes and for both products and GOBMs, deviations from independent observations, quantified with uRMSE, are at least 1 order of magnitude larger than maximum expected fCO_2 signals in recent mCDR modeling studies focused on hypothetical ocean alkalinity enhancement (OAE) deployments (Wang *et al* 2022, Zhou *et al* 2024). At present, approaches for monitoring, reporting and verifying (MRV) of mCDR project remain in development, with approaches, minimum detection limits, required monitoring durations, and many other details yet to be agreed (NASEM 2022). Thus, it remains unclear how the GOBMs and observation-based products evaluated in this manuscript may be included in MRV approaches. Since the air-sea gradient of fCO_2 is the dominant driver of air-sea CO_2 fluxes (McKinley *et al* 2017), it is clear that surface ocean fCO_2 will be a key variable in MRV systems (Subhas *et al* 2025), motivating this variable as our focus here.

This study demonstrates that significant improvements would be required before the current generation of observation-based products and GOBMs could be directly applied as a counterfactual for the ocean carbon sink without mCDR, and thus for mCDR additionality to be determined using these estimates as a reference state. It is possible that, despite their lack of precision, GOBMs could be used to assess mCDR additionality by using control minus control with mCDR perturbation simulations (Wang *et al* 2022, Zhou *et al* 2024). To build confidence in model-based mCDR additionality assessments, it will be important to demonstrate that model-estimated mCDR impacts are insensitive to the uncertainties and biases in the GOBMs revealed by this study and elsewhere (Fay and McKinley 2021, DeVries *et al* 2023, McKinley *et al* 2023, Terhaar *et al* 2024). It is also important to understand the extent to which observation-based products are skillful at local scales, since these products could be proposed as the basis for GOBM evaluation.

Across the superbiomes, we find substantial biases in 2000–2023 long-term mean fCO_2. Biases are substantially larger in GOBMs than in the products (figure 4), and with a much larger ensemble spread. This is consistent with much larger spread in the mean ocean carbon sink estimated by the GOBMs compared to the products (Fay and McKinley 2021, Terhaar *et al* 2024). Both superbiomes of the Northern Hemisphere have positive ensemble mean GOBM fCO_2 biases greater than +10 μatm against GLODAP+LDEOnoSOCAT (figure 4) and smaller, but still positive against the merged independent dataset that includes SOCAT data (figure S3). Three GOBMs (ACCESS, MPIOM-HAMOCC, NEMO-PlankTOM) do not include the water vapor correction in their treatment of atmospheric CO_2 (Friedlingstein *et al* 2026a), which should result in higher surface ocean fCO_2 (Fay *et al* 2021), and may partially explain positive biases in these models. Most GOBMs do not include fluxes of natural carbon from the land to the ocean via rivers (Regnier *et al* 2022) that should raise fCO_2 and cause CO_2 outgassing. Since this process is weak or missing in these GOBMs (DeVries *et al* 2023), GOBM underestimation of surface ocean fCO_2 could be expected. Instead, we find predominantly overestimation (figures 4 and S2).

The recent RECCAP2 assessment also found significant discrepancies in the GOBMs compared to the products (DeVries *et al* 2023, Terhaar *et al* 2024). Terhaar *et al* (2024) attribute GOBM biases to model spinup, ocean circulation, and chemistry errors. Many members of the CMIP5 Earth System Models (ESM) suite underestimate carbon uptake in the North Atlantic, with seasonal mechanisms being critical (Goris *et al* 2018). CMIP6 ESMs have substantial mean state discrepancies for the historical period compared to a similar suite of products. These errors are largely preserved in future projections and are responsible for at least 50% of the projection uncertainty with respect to the ocean carbon sink (McKinley *et al* 2023). This study adds to this accumulating evidence that GOBMs, and their closely-related ESM cousins, overestimate surface ocean fCO_2 and thus underestimate the mean ocean carbon sink. Given the mounting evidence of substantial GOBM errors, a recent publication (Friedlingstein *et al* 2026b) proposes that the Global Carbon Budget adjust the ocean carbon sink estimated by GOBMs: essentially increasing them by 10%. This adjustment is used in the most recent Global Carbon Budget release Friedlingstein *et al* 2026a, McKinley 2026).

There are a handful of ocean biogeochemical models that use data assimilation in an attempt to increase fidelity. In the ECCO family of models, the adjoint method is used to adjust the physical model state and then a Green's Function approach is used with data such as GLODAP and SOCAT to adjust parameters of the biogeochemical module (Carroll *et al* 2020, 2022, Moseley *et al* 2026). Could such data

assimilation efforts substantially improve these comparisons shown here? In section S4, we extend these comparisons to two models in the ECCO family and do not find significant improvements.

Observation-based products are generally more skillful than GOBMs, but also have substantial uncertainties that need to be addressed with additional data collection and algorithmic improvements (Heimdal *et al* 2026). For example, uRMSEs are 37 μatm in the N.SS superbiome, and at least 20 μatm in other superbiomes. Non-seasonal variability is smaller magnitude and poorly represented (figure 5).

Despite mean state and seasonal discrepancies, both products and GOBMs consistently estimate the trend in fCO_2 in all superbiomes. This is consistent with the very strong forced signal from rising atmospheric CO_2 concentrations (McKinley *et al* 2020, DeVries *et al* 2023, Gruber *et al* 2023).

In this study, sparse observations across large open-ocean superbiomes are aggregated in this study to timeseries as the basis for comparison to products and models. This approach is justified by the common physical-biogeochemical states of these regions (Fay and McKinley 2014) and has been widely applied (Fay and McKinley 2021, Rodgers *et al* 2023, Terhaar *et al* 2024, Müller *et al* 2025). Nonetheless, aggregation has the potential to interweave temporal signals and those arising from heterogeneous spatial sampling. This issue does not impact biome-scale estimation of bias, uRMSE and correlation for the full timeseries ('All'), but does have potential to impact interpretation of the relative importance of time-varying seasonal mechanisms vs. variability at higher or lower temporal frequencies as the driver of underlying model or product errors. In the Supplementary Material, a comparison of the biome-aggregated timeseries of the N.STPS biome to the timeseries at Bermuda Atlantic Timeseries (BATS) and Hawai'i Ocean Timeseries (HOT) indicates that spatial averaging does impact quantitative results for uRMSE and bias, but does not change qualitative results with respect to substantially greater skill for seasonal phasing (modest to high correlations) than for variability (section S5).

Outside the subtropics, seasonality phase is much better represented in the products (modest to high correlations) than in the GOBMs (low correlations). Nonetheless, high uRMSEs persist in both products and GOBMs on the seasonal timescale, particularly at high latitudes (figures 3 and 5). Prior studies that have attributed seasonal errors in GOBM fCO_2 to errors in the representation of physical and biological processes associated with dissolved inorganic carbon (DIC) and alkalinity cycles (Mongwe *et al* 2018, DeVries *et al* 2023, Rodgers *et al* 2023, Bushinsky *et al* 2026). As shown here and also by Gloege *et al* (2021), the products offer more reliable estimates of seasonal amplitude and phase for fCO_2. Thus, the products could be useful in guiding future GOBM development.

Errors in non-seasonal variability are relatively small in terms of uRMSEs. The phasing of this variability is poorly estimated by both products and GOBMs, with the exception of the products in the equatorial superbiome where the ensemble mean correlation is 0.6 (figure 5). Given the sparsity of the independent data (table 2), we do not attempt to determine the extent to which non-seasonal variability is due to long-term variability as opposed to high-frequency events such as storms, fronts and eddies. Sustained observations that can better constrain non-seasonal variability will be needed to make future progress in this area.

In this study, we compare observation-based products and GOBMs to the same independent dataset. Though these independent data are sparse, they are global in scope and fully independent of the products. While it is well known that the broad agreement across models and products for the global ocean carbon sink masks significant smaller-scale uncertainties (McKinley *et al* 2016, 2020, Fay and McKinley 2021, Gloege *et al* 2021, DeVries *et al* 2023, Hauck *et al* 2023, Roobaert *et al* 2026), this study explicitly quantifies these uncertainties at the local scale for the first time. We find that the products are substantially more skillful than the GOBMs, with lower bias, lower uRMSEs and higher correlations to independent data. Biome-aggregated seasonal signals are dominant and non-seasonal variability is of smaller magnitude and poorly represented. A key future direction is to improve the representation of fCO_2 seasonality in GOBMs and to collect more surface ocean fCO_2 data to both improve observation-based products and to expand independent datasets for more comprehensive skill assessments.

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Data availability statement

The observation-based products and GOBMs from the Global Carbon Budget 2024 are available at https://zenodo.org/records/14092497. The LDEO-Residual product is available at https://oceancarbon.ldeo.columbia.edu. GLODAP is available from https://glodap.info/index.php/merged-and-adjusted-data-product-v2-2023/. LDEO pCO_2 database is available via NCEI at www.ncei.noaa.gov/access/ocean-carbon-acidification-data-system/oceans/LDEO_Underway_Database/. SOCATv2024 database is available at https://socat.info/index.php/version-2024/.

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